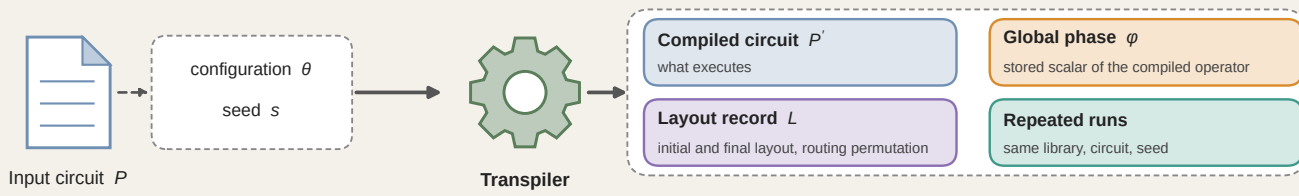


1 A Transpilation returns more than a circuit

$$C(P, \theta, s) = (P', L, \varphi)$$

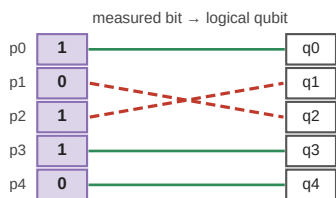


2 What each oracle can observe

Property	Output-equivalence oracle	Matched oracle (this work)
Compiled circuit P'	✓ compares operator, state or distribution with the input	output faults are the scope of the output oracle
Layout record L	✗ may miss it: invisible if the operator does not depend on it	✓ Contract checker (K0–K2) MR-1 permutation consistency
Global phase φ	✗ cannot observe: global phase is normalized away by definition	✓ Global-phase tracker (exact $n \leq 12$, sampled 13–22)
Repeated runs	✗ cannot observe: judges each run on its own	✓ Determinism runner (raw and functional fingerprints)

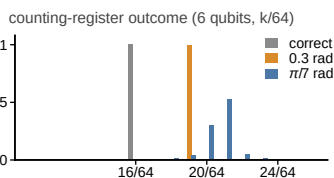
3 Consequence for a quantum computation when the property is corrupted

Layout record (readout test)



30 of 60 random circuits decoded wrongly

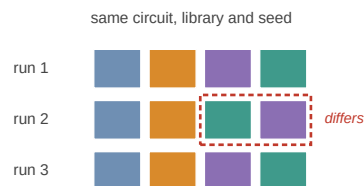
Global phase (phase estimation)



output equivalence still passes

$P(\text{true outcome}) 1.00 \rightarrow 3.3 \times 10^{-4}$ (0.3 rad)
and 4.7×10^{-3} ($\pi/7$ rad)

Reproducibility (schematic)



two commuting instructions swap order, the executed output is identical

replication and debugging suffer